



ECEN 314 - Signals and Systems

Scott L. Miller

Module 3 - Signals in the Time Domain

Learning Objectives



- Be familiar with the form of some common signals.
- Understand what a delta function is and some of its basic properties.
- Be able to perform arithmetic operations on signals.
- Be able to form basic operations on signals, most importantly, flipping, shifting and scaling.
- Be able to decompose a signal into even and odd parts.

Some Common Signals



Sine Wave (sinusoid)

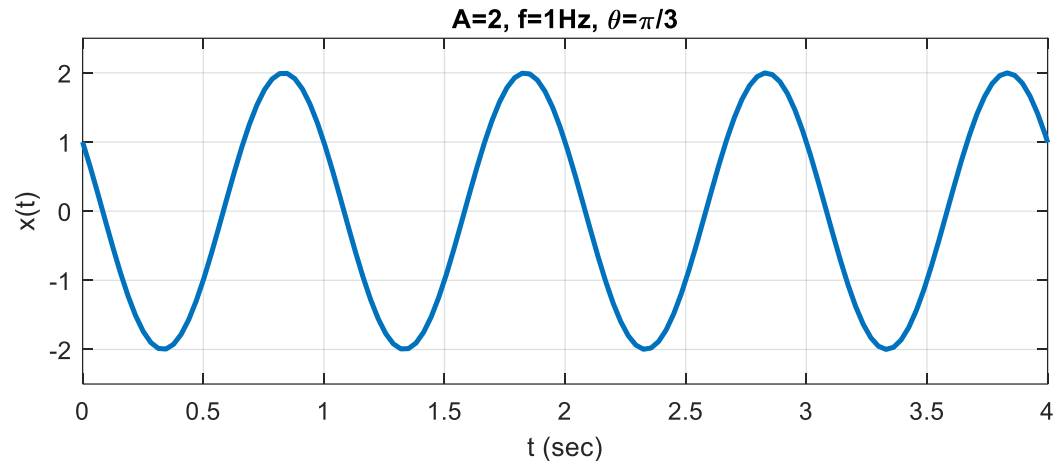
$$x(t) = A\cos(\omega t + \theta)$$

A = Amplitude

ω = frequency (in rad./sec)

$\omega = 2\pi f$, f = frequency (in Hz)

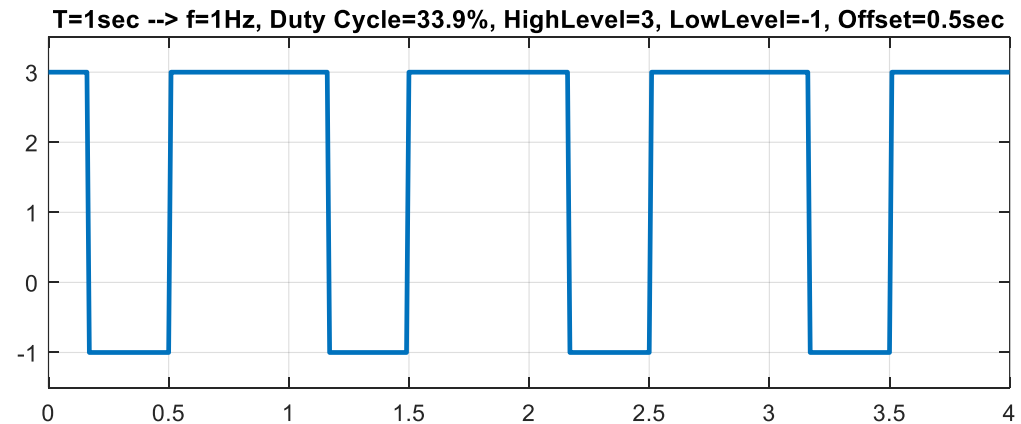
θ = phase (in radians)



Square-Wave

Described by:

- ✓ T = period
- ✓ $f = \frac{1}{T}$ = frequency (in Hz)
- ✓ Duty cycle = % of time in high state
- ✓ High level
- ✓ Low level
- ✓ Offset

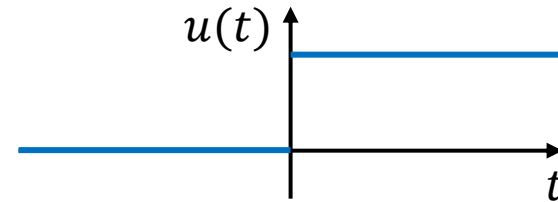


Some Common Signals



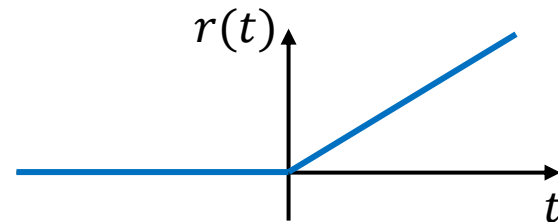
Unit Step Function

$$u(t) = \begin{cases} 1, & t > 0, \\ 0, & t < 0. \end{cases}$$



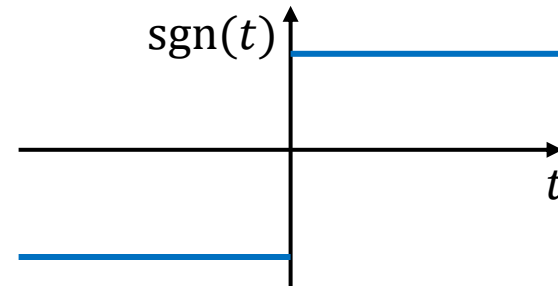
Unit Ramp Function

$$r(t) = \int_{-\infty}^t u(\tau) d\tau = \begin{cases} t, & t \geq 0, \\ 0, & t < 0. \end{cases}$$



Signum (sign) Function

$$\text{sgn}(t) = \begin{cases} 1, & t > 0, \\ -1, & t < 0. \end{cases}$$



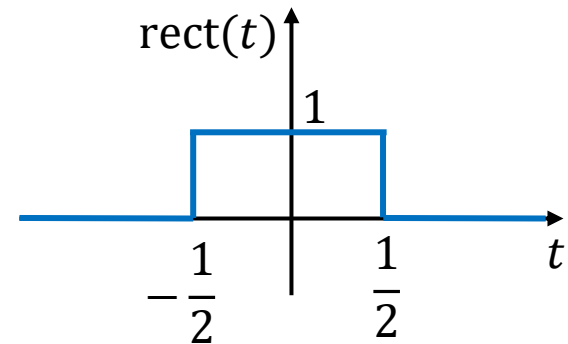
Some Common Signals



Rectangle Function

$$\text{rect}(t) = \begin{cases} 1, & |t| < 1/2, \\ 0, & |t| > 1/2. \end{cases}$$

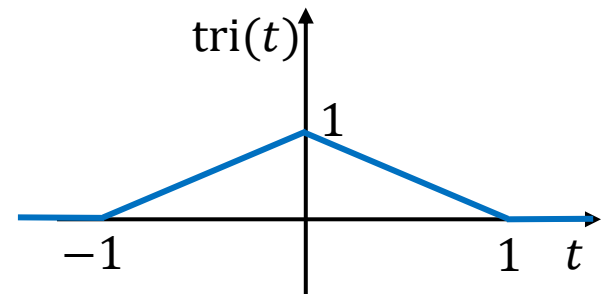
Note: We are defining the rectangle function as a square pulse of width 1, height 1, centered at $t = 0$.



Triangle Function

$$\text{tri}(t) = \begin{cases} 1 - |t|, & |t| < 1, \\ 0, & |t| > 1. \end{cases}$$

Note: The triangle has a height of 1 but a width of 2. It will become clear later in the course why we defined it that way.

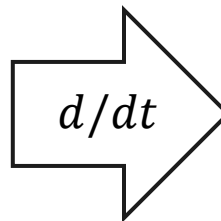
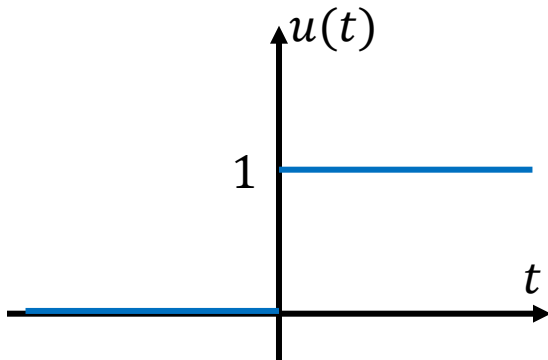


Delta Functions

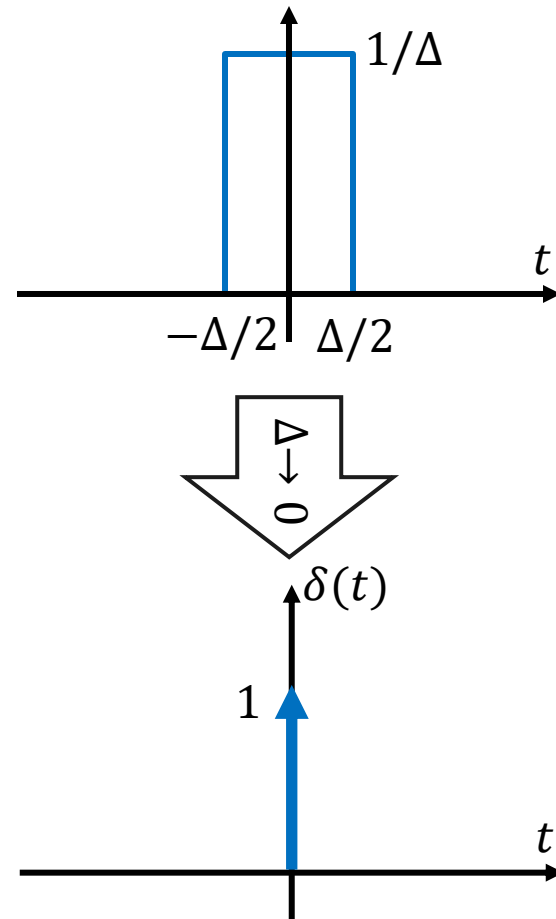


One of the waveforms that plays a fundamental role in the study of signals and systems is the *Dirac Delta Function* a.k.a. the *unit impulse function*, $\delta(t)$.

Another way to view a delta function is as the derivative of the unit step function.



One way to view the delta function is as an infinitely tall, infinitely narrow pulse with a unit area underneath.



Delta Functions



One of the important properties of delta functions is known as the *sifting property*, which says that for some arbitrary function $x(t)$

$$\int_a^b \delta(t - t_o)x(t)dt = \begin{cases} x(t_o), & \text{if } a < t_o < b, \\ 0, & \text{otherwise.} \end{cases}$$

Proof:

Using the $\delta(t - t_o) = \frac{d}{dt}u(t - t_o)$ interpretation:

$$\begin{aligned} & \int_a^b \frac{d}{dt}u(t - t_o)x(t)dt \\ &= x(t) u(t - t_o) \Big|_a^b - \int_a^b u(t - t_o) \frac{dx}{dt} dt \end{aligned}$$

Assuming that $a < t_o < b$,

$$\begin{aligned} &= x(b) - \int_{t_o}^b \frac{dx}{dt} dt \\ &= x(b) - [x(b) - x(t_o)] \\ &= x(t_o) \end{aligned}$$

Proof: Using the tall thin pulse interpretation:

The integrand $\delta(t - t_o)x(t)$ is 0 everywhere except at $t = t_o$ and is equivalent to $\delta(t - t_o)x(t_o)$ so that

$$\begin{aligned} & \int_a^b \delta(t - t_o)x(t)dt \\ &= \int_a^b \delta(t - t_o)x(t_o)dt \\ &= x(t_o) \int_a^b \delta(t - t_o)dt = x(t_o) \end{aligned}$$

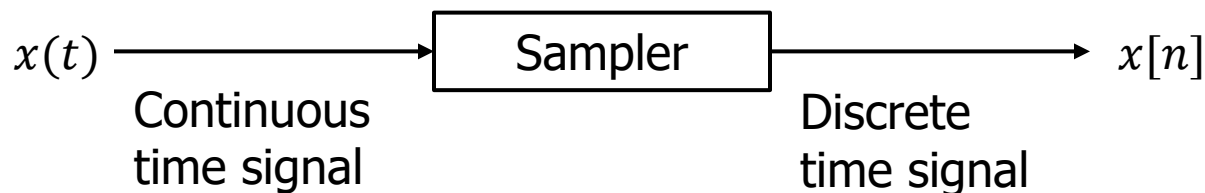
Since the area underneath the delta function is one.

Signals in Discrete Time



Many times, it is helpful/necessary to work with functions of time (signals) where the time variable is discrete in nature. This happens, for example, when we want to process signals with computers/software.

Discrete time signals result from taking periodically spaced samples of continuous time signals. This is the first step in the process of analog to digital conversion.



To help us clarify whether a signal we are working with is in continuous or discrete time, we will use parentheses $()$ for continuous time and square brackets $[]$ for discrete time.

$y(t) \rightarrow$ the signal y is a function of a continuous variable t . That is $t \in (-\infty, \infty)$.

$z[n] \rightarrow$ the signal z is a function of the discrete variable n . That is

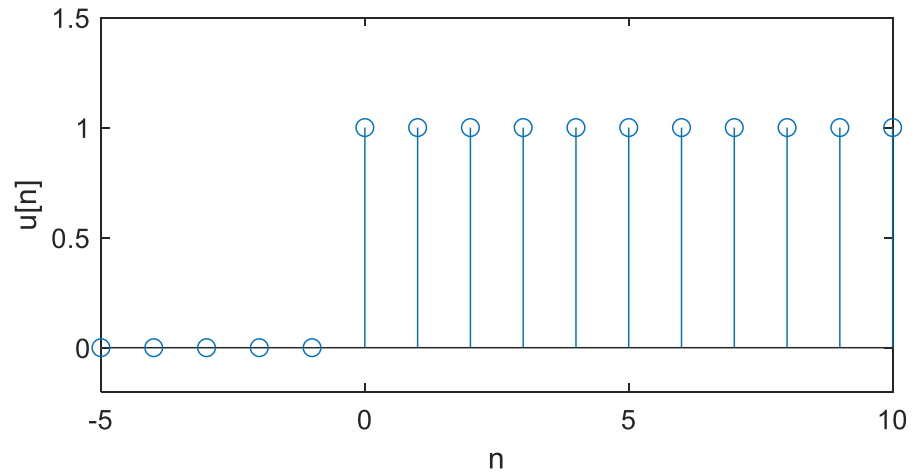
$$n \in \dots, -3, -2, -1, 0, 1, 2, 3 \dots$$

Some Signals in Discrete Time



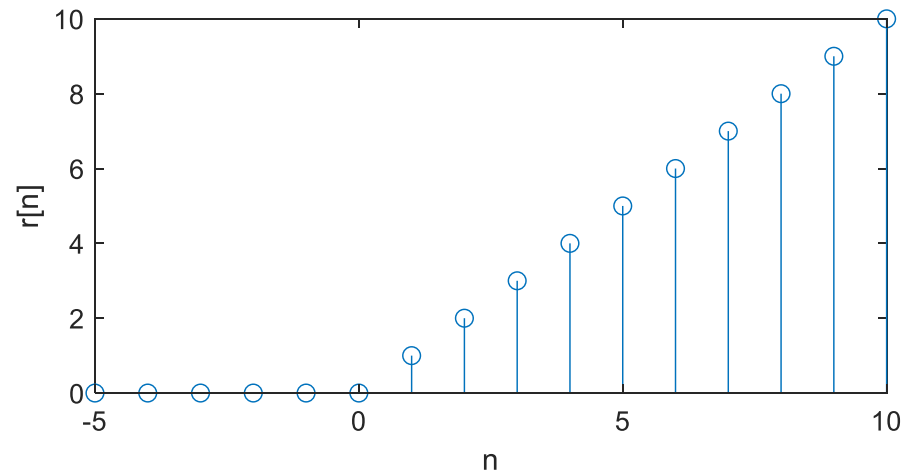
Unit Step Function

$$u[n] = \begin{cases} 1, & n \geq 0, \\ 0, & n < 0. \end{cases}$$



Unit Ramp Function

$$r[n] = \begin{cases} n, & n \geq 0, \\ 0, & n < 0. \end{cases}$$



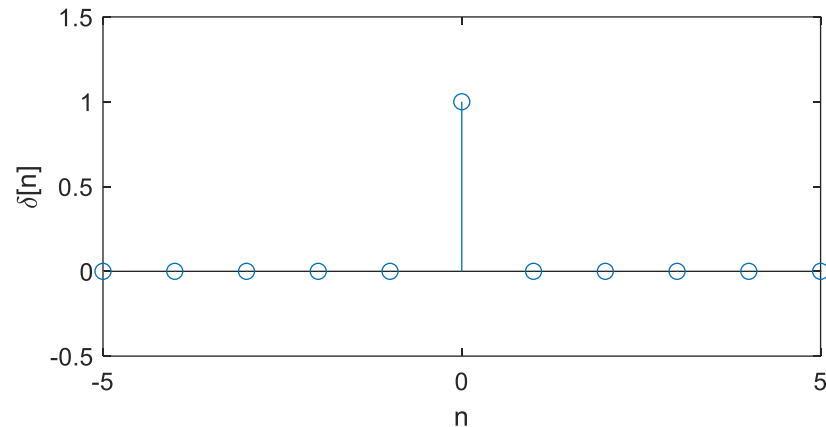
Some Signals in Discrete Time



Delta Function

$$\delta[n] = \begin{cases} 1, & n = 0, \\ 0, & n \neq 0. \end{cases}$$

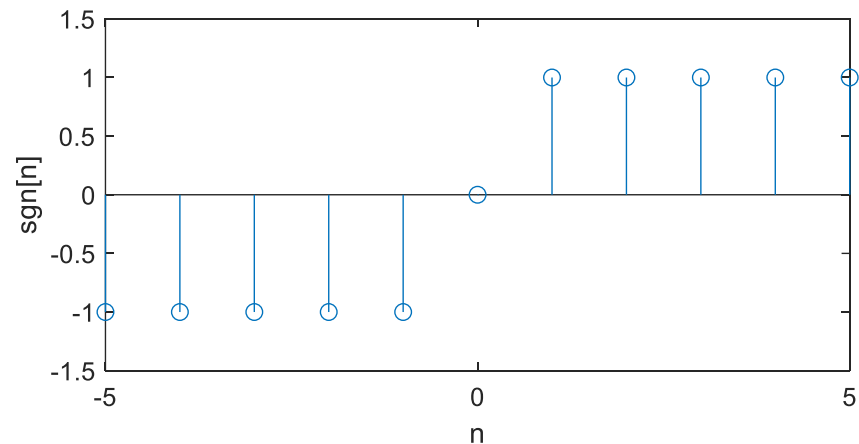
Note: For the discrete time delta function, the value of the function at $n = 0$ is not infinity (it is 1).



Signum (sign) Function

$$\text{sgn}[n] = \begin{cases} 1, & n > 0, \\ 0, & n = 0, \\ -1, & n < 0. \end{cases}$$

Note: For the discrete time signum function, the value of the function at $n = 0$ is 0.



Signals in Discrete Time



Example 3.1: Sketch each of the following signals.

(a) $x[n] = \left(\frac{1}{2}\right)^n u[n],$

(b) $y[n] = \cos\left[\frac{\pi n}{4}\right].$

Arithmetic With Signals

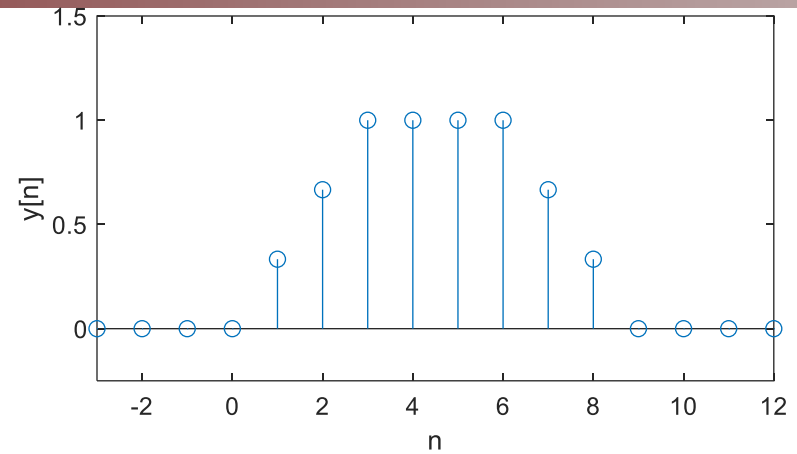
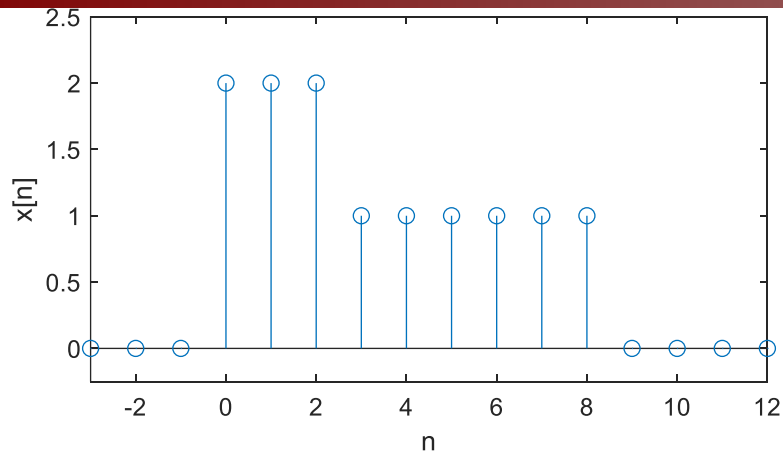


We will often need to perform various operations involving signals. To start with we will do basic arithmetic: addition, subtraction, multiplication, and maybe on rare occasions division.

It is particularly easy to do arithmetic with discrete time signals as we can just perform the operation on a point-by-point basis.

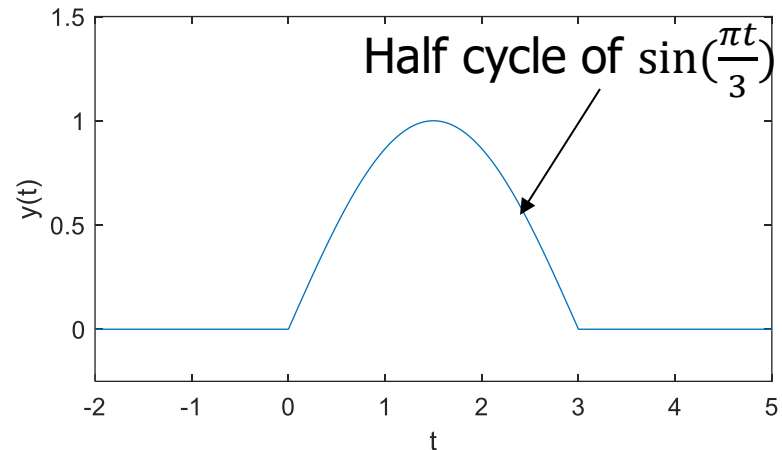
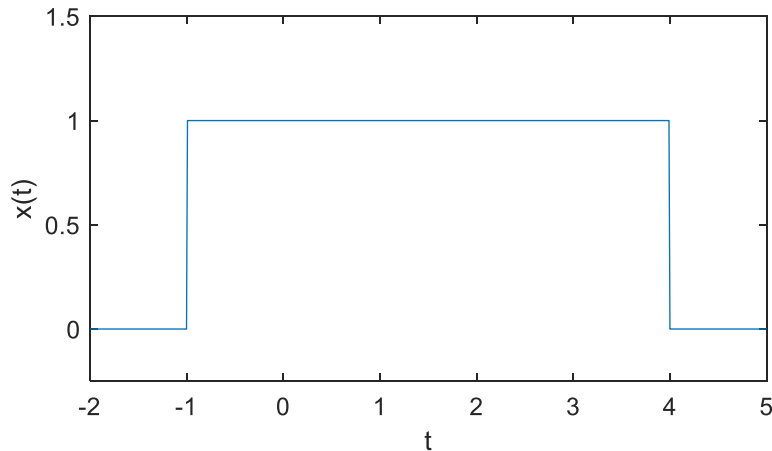
Example 3.2: Sketch $x[n] = u[n] + r[n]$ and $x(t) = u(t) + r(t)$.

Arithmetic With Signals



Example 3.3: Sketch $w[n] = x[n] - y[n]$ and $z[n] = x[n] \cdot y[n]$ for the discrete time signals $x[n]$ and $y[n]$ shown above.

Arithmetic With Signals



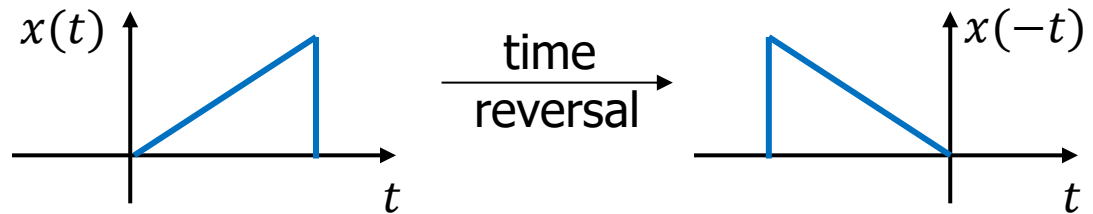
Example 3.4: Sketch $w(t) = x(t) - y(t)$ and $z(t) = x(t) \cdot y(t)$ for the continuous time signals $x(t)$ and $y(t)$ shown above.

Operations on Signals



Time reversal

$$x(t) \rightarrow x(-t)$$

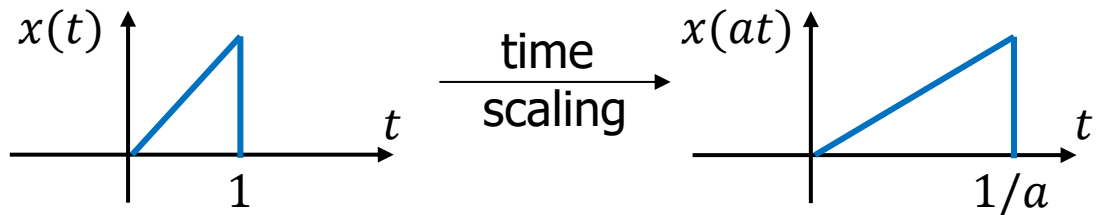


Time scaling

$$x(t) \rightarrow x(at)$$

$0 < a < 1 \rightarrow$ expand in time

$a > 1 \rightarrow$ compress in time

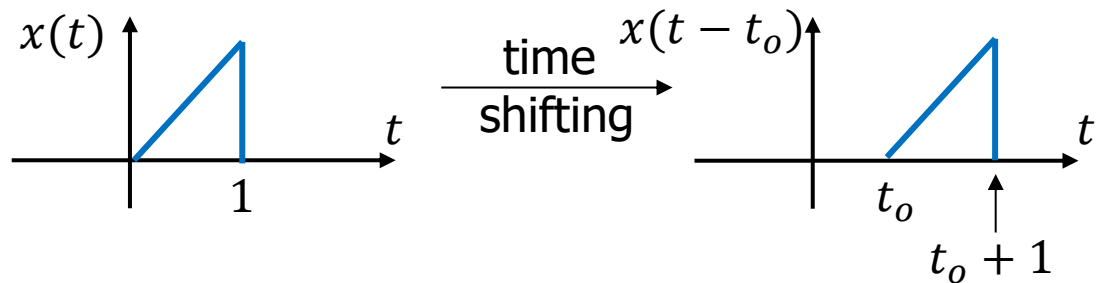


Time shifting

$$x(t) \rightarrow x(t - t_o)$$

$t_o > 0 \rightarrow$ shift right

$t_o < 0 \rightarrow$ shift left



Operations on Signals



Example 3.5: Sketch each of the following continuous time signals

- a) $\text{rect}(t/6)$
- b) $\text{rect}(t + 2)$
- c) $\text{rect}(3t + 2)$
- d) $\text{rect}(2 - t/2)$

Operations on Signals



Given a signal $x(t)$, a time transformation of the form $x(at - b)$ can be written as one of two equivalent two step operations.

Shift then scale

Step 1: Shift by b : $x(t) \rightarrow x(t - b)$

Step 2: Scale by a : $x(t - b) \rightarrow x(at - b)$

Scale then shift

Step 1: Scale by a : $x(t) \rightarrow x(at)$

Step 2: Shift by b/a :

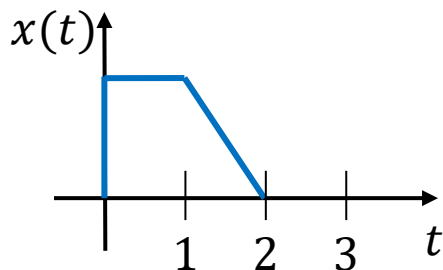
$$x(at) \rightarrow x\left(a\left(t - \frac{b}{a}\right)\right) = x(at - b)$$

Example 3.6:

Suppose $x(t)$ is as shown.

(a) Sketch $x(2t - 1)$. Try it both ways.

(b) Then sketch $x(1 - 2t)$.



Operations on Signals



Even and Odd Functions



A function $x(t)$ is even if $x(t) = x(-t)$.

→ If you flip it horizontally (time reverse it) you get the same thing.

A function $x(t)$ is odd if $x(t) = -x(-t)$.

→ If you flip it horizontally (time reverse it) and vertically (negate it) you get the same thing.

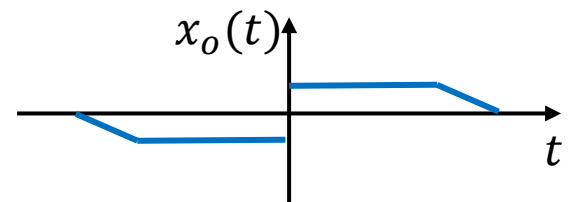
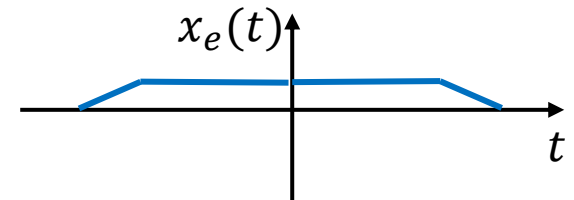
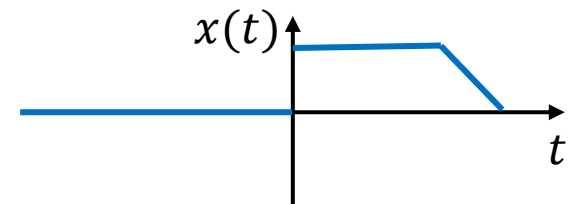
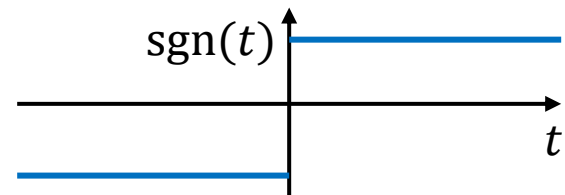
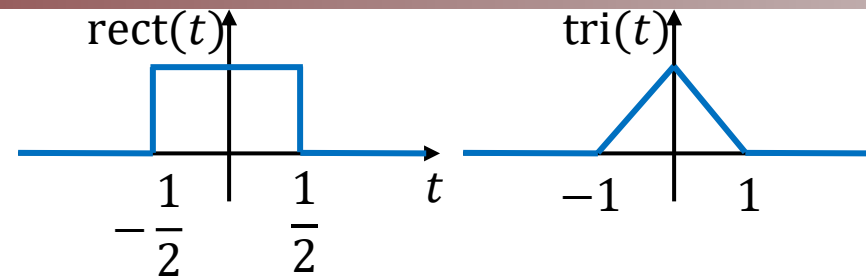
For any function $x(t)$ that is not necessarily either even or odd, note that

✓ $x_e(t) = \frac{x(t) + x(-t)}{2}$ is even,

✓ $x_o(t) = \frac{x(t) - x(-t)}{2}$ is odd,

✓ $x(t) = x_e(t) + x_o(t)$.

→ Any signal can be decomposed into the sum of an even part and an odd part.



Even and Odd Functions



Example 3.7: Decompose the signals shown into and even and odd parts.

